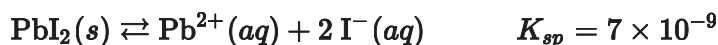


Unit 7 Progress Check: FRQ

Name _____

1. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.



The dissolution of $\text{PbI}_2(s)$ is represented above.

- (a) Write a mathematical expression that can be used to determine the value of S , the molar solubility of $\text{PbI}_2(s)$. (Do not do any numerical calculations.)



Please respond on separate paper, following directions from your teacher.

- (b) If $\text{PbI}_2(s)$ is dissolved in $1.0 \text{ M NaI}(aq)$, is the maximum possible concentration of $\text{Pb}^{2+}(aq)$ in the solution greater than, less than, or equal to the concentration of $\text{Pb}^{2+}(aq)$ in the solution in part (a)? Explain.



Please respond on separate paper, following directions from your teacher.

Compound	K_{sp}
PbCl_2	2×10^{-5}
PbI_2	7×10^{-9}
$\text{Pb}(\text{IO}_3)_2$	3×10^{-13}

- (c) A table showing K_{sp} values for several lead compounds is given above. A saturated solution of which of the compounds has the greatest molar concentration of $\text{Pb}^{2+}(aq)$? Explain. (Do not do any numerical calculations.)



Please respond on separate paper, following directions from your teacher.

2. For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

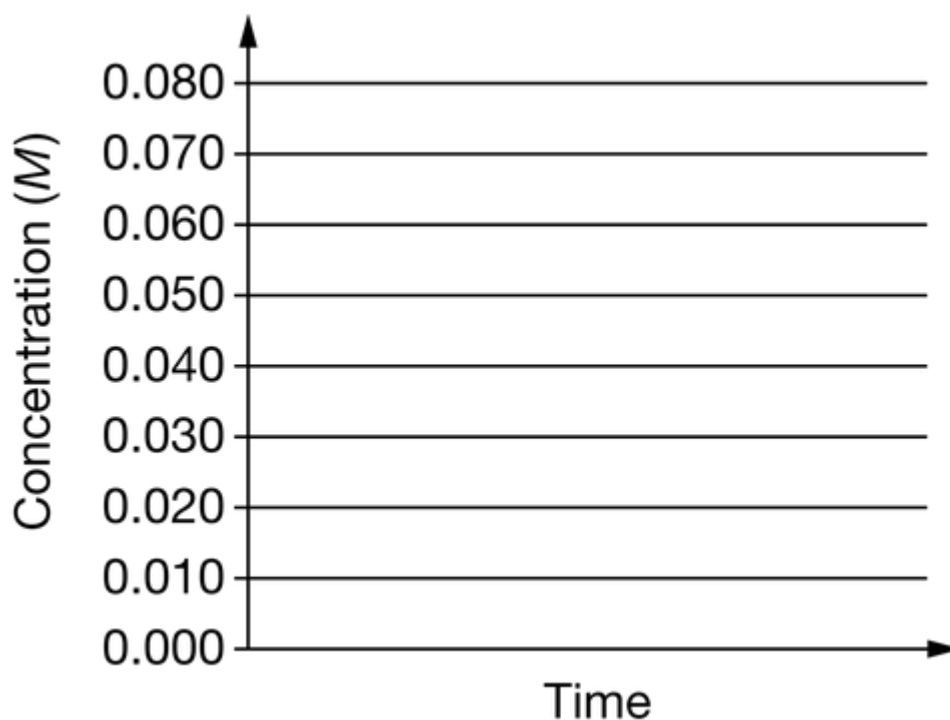


Unit 7 Progress Check: FRQ

Gas	Initial Concentration (M)
H_2	0.030
I_2	0.015
HI	?

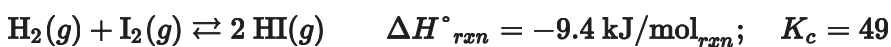
Samples of three gases, $H_2(g)$, $I_2(g)$, and $HI(g)$, were combined in a rigid vessel. The initial concentrations of $H_2(g)$ and $I_2(g)$ are given in the table above.

(a) The original value of the reaction quotient, Q_c , for the reaction of $H_2(g)$ and $I_2(g)$ to form $HI(g)$ (before any reactions take place and before equilibrium is established), was **5.56**. On the following graph, plot the points representing the initial concentrations of all three gases. Label each point with the formula of the gas.



Please respond on separate paper, following directions from your teacher.

Equilibrium was established at a certain temperature according to the following chemical equation.



After equilibrium was established, the concentration of $H_2(g)$ was **0.020 M**.



Unit 7 Progress Check: FRQ

(b) On the graph above, carefully draw three curves, one for each of the three gases, starting from the initial points you drew in part (a). The curves must show how the concentration of each of the three gases changed as equilibrium was established.



Please respond on separate paper, following directions from your teacher.

$\text{H}_2(g)$, $\text{I}_2(g)$, and $\text{HI}(g)$ are at equilibrium at a different temperature in a different vessel.

(c) When the temperature in the vessel is decreased, does the equilibrium shift to the right, favoring the product, or to the left, favoring the reactants? Justify your answer.



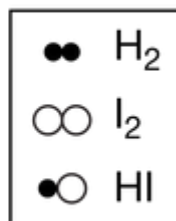
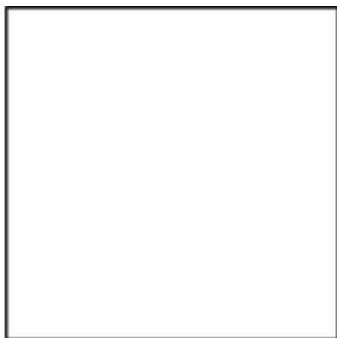
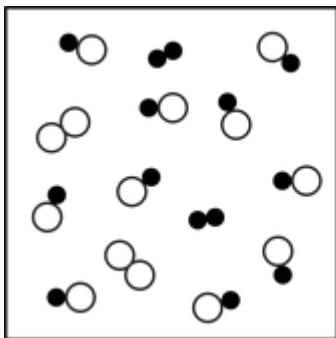
Please respond on separate paper, following directions from your teacher.

(d) Does the value of K_c increase, decrease, or remain the same when the temperature is decreased? Justify your answer based on the expression for K_c and the concentrations of the product and reactants.



Please respond on separate paper, following directions from your teacher.

(e) In the following empty box, draw an appropriate number of each type of molecule to represent a possible new equilibrium at the lower temperature.



Please respond on separate paper, following directions from your teacher.